

Material Selection System

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July 28, 2026

Abstract

Selecting the most suitable engineering material for a given application is a critical task that depends on multiple factors such as mechanical properties, environmental conditions, manufacturing constraints, cost, and performance requirements. This project presents a Database Management System (DBMS) based Material Selection Recommendation System that assists users in identifying the most appropriate material for their specific applications.

The system allows users to provide details such as the intended application, operating conditions, load requirements, temperature range, corrosion resistance, weight constraints, manufacturing process, and budget. Based on these inputs, the application queries a structured material database containing comprehensive information on metals, polymers, ceramics, composites, and other engineering materials. Using predefined selection criteria and database queries, the system recommends the most suitable materials along with their key properties and justifications.

The proposed system simplifies the material selection process, reduces decision-making time, minimizes human error, and serves as an effective decision-support tool for students, engineers, researchers, and designers.

The *Material Selection Recommendation System* is a user-friendly database application developed using DBMS concepts to streamline the process of selecting engineering materials. Users specify their design requirements, including working environment, mechanical loads, thermal conditions, corrosion resistance, manufacturing constraints, and cost limitations. The system stores detailed information about various engineering materials and retrieves the best matches using efficient database queries.

Unlike conventional material handbooks, which require extensive manual searching and comparison, this system automates the selection process by filtering materials based on user-defined constraints. The centralized database ensures data consistency, easy maintenance, and scalability, allowing new materials and properties to be incorporated with minimal effort. The project demonstrates the practical application of database management techniques in engineering decision-making while providing an accurate, efficient, and accessible material recommendation platform.

Submitted on: July 28, 2026 at 15:00